Supplemental Document

Optics EXPRESS

Axial profiling of interferometric scattering enables an accurate determination of nanoparticle size: supplement

KATEŘINA ŽAMBOCHOVÁ,[1,2] IL-BUEM LEE,[1] JIN-SUNG PARK,[1] SEOK-CHEOL HONG,[1,3,5] AND MINHAENG CHO[1,4,6,]

[1]*Center for Molecular Spectroscopy and Dynamics, Institute for Basic Science (IBS), Seoul 02841, Republic of Korea*
[2]*Department of Natural Sciences, Faculty of Biomedical Engineering, Czech Technical University in Prague, Kladno, 272 01, Czech Republic*
[3]*Department of Physics, Korea University, Seoul 02841, Republic of Korea*
[4]*Department of Chemistry, Korea University, Seoul 02841, Republic of Korea*
[5]*hongsc@korea.ac.kr*
[6]*mcho@korea.ac.kr*

This supplement published with Optica Publishing Group on 6 March 2023 by The Authors under the terms of the Creative Commons Attribution 4.0 License in the format provided by the authors and unedited. Further distribution of this work must maintain attribution to the author(s) and the published article's title, journal citation, and DOI.

Supplement DOI: https://doi.org/10.6084/m9.figshare.22047950

Parent Article DOI: https://doi.org/10.1364/OE.480337

SUPPLEMENTARY INFORMATION

SI contents

Fig. S1. Schematic diagram of the F-iSCAT setup and data acquisition by z-scanning 2
Vectorial point spread function modeling of nanoparticle image by iSCAT microscopy ..3
Fig. S2. Illustration of the optical components of the iSCAT microscopy 3
A. Derivation of the reference and scattered fields in the pupil region of the imaging lens 3
B. Electric fields at the focal region of the imaging lens derived from the Richards-Wolf integral.....4
Figs. S3-4. Simulations of the axial intensity profile.....6
Fig. S5. Finding the best fit to an experimental axial profile with given n_i and z_p via RMSE evaluation 7
Fig. S6. RMSE evaluation to determine the model parameters, n_i and z_p.....8
Fig. S7. Representative results of PSF modeling analysis for PS and latex NPs 9
Fig. S8. Representative results of PSF modeling analysis for fNDs 10
References.....11

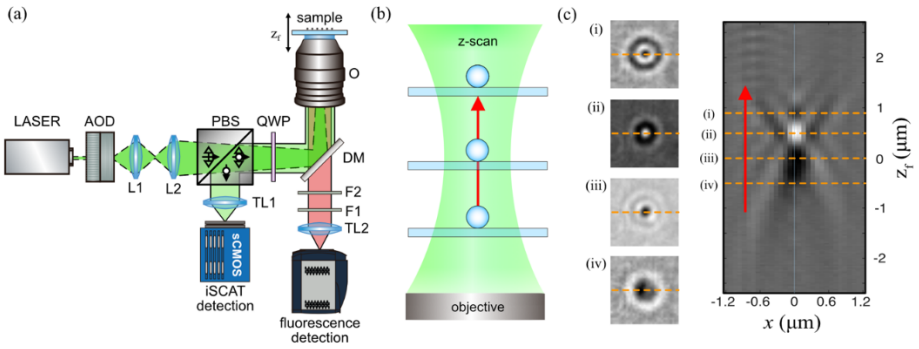


Fig. S1. Schematic diagrams of the F-iSCAT setup and data acquisition by z-scanning. (a) Fluorescence combined iSCAT microscope. The outlined and dashed-outlined green beams mark the scattered and illuminating lights, respectively, and the red beam shows fluorescence from NPs. Optical elements in the setup are acousto-optic deflector (AOD), lenses (L1, L2, TL1, TL2), polarizing beam splitter (PBS), quarter-wave plate (QWP), objective (O), dichroic mirror (DM), notch filter (F1), and emission filter (F2). (b-c) Illustration of z-scanning and representative axial variation of iSCAT image by an NP (dia.: 120 nm). (b) Sample is scanned along the z-axis (red arrow) over a range of 5 μm around the focal plane by moving the sample stage. (c) Experimental SHIP image by stacking the cross-sectional intensity profile (guided on the orange dashed lines) along the diameter of the NP image (indicated by orange-dashed lines in (i-iv)).

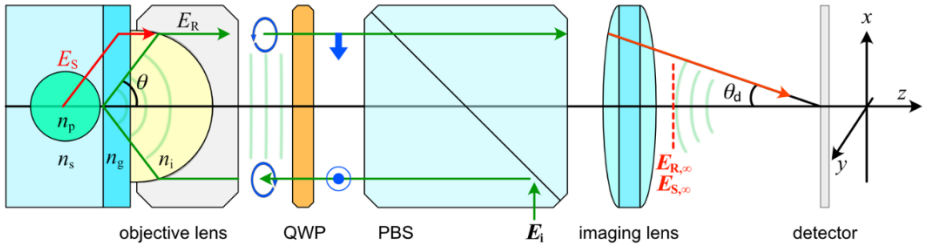


Fig. S2. Illustration of the optical components of the iSCAT microscopy.

A generalized Jones matrix describes the effect of an optical element on ray propagation. Our system consists of a polarizing beam splitter (PBS), a quarter-wave plate (QWP), several lenses, and transmitting/reflecting interfaces for which we used the relevant Jones matrices [1]. The matrices were applied in order to trace the polarization state of light. We consider the incident beam that impinges on the PBS. By the PBS, the beam becomes 45° linearly polarized with respect to the optic axis of the subsequent QWP and circularly polarized at the back focal plane of the objective by the QWP. The beam shines at the sample through the objective lens, and the reflected and scattered fields propagate all the way to the dotted line after the imaging lens as illustrated in Figure S2, passing through the objective lens, QWP, PBS, and the imaging lens. The Richards-Wolf integral is then applied to those fields to find out the field at the focal region (detector).

A. Derivation of the reference and scattered fields in the pupil region of the imaging lens

Reference field: The reference field E_R is the light reflected at the glass and water/sample interface. The reference field at the pupil of the imaging lens ($\vec{E}_{R,\infty}$) is given as:

$$\vec{E}_{R,\infty}(\theta_d,\phi) = \mathbb{R}_z^{-1}(\phi)\cdot\mathbb{L}(\theta_d)\cdot\mathbb{R}_z(\phi)\cdot\mathbb{P}(0°)\cdot\mathbb{W}(90°,45°)\cdot\mathbb{R}_z^{-1}(\phi) \cdot\mathbb{F}_R\cdot\mathbb{R}_z(\phi)\cdot\mathbb{W}(90°,45°)\cdot\mathbb{P}(90°)\cdot\vec{E}_i, \quad \text{(S1)}$$

where $\mathbb{P}(0°)$, $\mathbb{P}(90°)$, $\mathbb{W}(90°,45°)$, $\mathbb{R}_z(\phi)$, $\mathbb{L}(\theta_d)$, and $\mathbb{F}_R$ are the Jones matrices for polarizing beam splitters for transmission and reflection, quarter-wave plate, coordinates rotation by ϕ about the optic axis for local p-/s-wave decomposition, imaging lens with the zenith angle θ_d, and Fresnel field transformation by reflection, respectively. The incident field is given by $\vec{E}_i = (E_x,E_y) = (0,1)$.

$$\vec{E}_{R,\infty}(\theta_d,\phi) = C_R\cdot i(r_p+r_s)\begin{bmatrix} -(1-\cos\theta_d)\cos\phi\sin\phi \\ \cos^2\phi + \sin^2\phi\cos\theta_d \end{bmatrix} = \frac{1}{2}C_R\cdot i(r_p+r_s)\begin{bmatrix} -(1-\cos\theta_d)\sin 2\phi \\ (1+\cos\theta_d)+(1-\cos\theta_d)\cos 2\phi \end{bmatrix}, \quad \text{(S2)}$$

where r_p and r_s are the Fresnel reflection coefficients for p- and s-polarized light and C_R is just the overall numerical pre-factor.

Scattered field: In this work, we only deal with scattering by a spherical particle. The scattered field by a spherical scatterer ($\vec{E}_s$) takes a complex form, as shown

below. The scattered field at the pupil of the imaging lens ($\vec{E}_{S,\infty}$) is expressed in terms of $\vec{E}_s$ defined in local spherical coordinates:

$$\vec{E}_{S,\infty}(\theta_d,\phi) = \mathbb{R}_z^{-1}(\phi).\mathbb{L}(\theta_d).\mathbb{R}_z(\phi).\mathbb{P}(0°).\mathbb{W}(90°,45°).\mathbb{R}_z^{-1}(\phi).\mathbb{F}_T.\vec{E}_s\,, \tag{S3}$$

where $\mathbb{F}_T$ is the Fresnel transmission matrix at the interface. If the incident field at the sample is circularly polarized, the scattered field is given by $\vec{E}_S = \hat{\theta}E_{S\theta} + i\hat{\phi}E_{S\phi}$. $\vec{E}_S$ is described in terms of the spherical coordinates (r,θ,ϕ) defined in the sample chamber with the origin at the center of the spherical particle. θ is the angle of a ray made with respect to the optic axis, and thus, it defines the polar direction of a ray in the sample space (immersion oil). The scattered field from a spherical scatterer is acquired by the Mie-scattering theory; $E_{S\theta}$ and $E_{S\phi}$, the polar and azimuthal components of the scattered electric field, respectively, shall be expanded as a linear combination of $\pi_n = \frac{P_n^1}{\sin\theta}$ and $\tau_n = \frac{dP_n^1}{d\theta}$

$$E_{S\theta} = \frac{1}{\sqrt{2}}\frac{e^{ikr}}{-ikr}e^{i\phi}S_2(\theta,\phi) \propto e^{i\phi}\sum\frac{2n+1}{n(n+1)}(a_n\tau_n + b_n\pi_n)\,, \tag{S4}$$

$$E_{S\phi} = \frac{1}{\sqrt{2}}\frac{e^{ikr}}{-ikr}e^{i\phi}S_1(\theta,\phi) \propto e^{i\phi}\sum\frac{2n+1}{n(n+1)}(a_n\pi_n + b_n\tau_n), \tag{S5}$$

where S_1 and S_2 are some functions defined in terms of π_n and τ_n, and a_n and b_n are the coefficients of π_n and τ_n [2].

The final form of the scattered field is simplified as follows:

$$\begin{aligned}\vec{E}_{S,\infty}(\theta_d,\phi) &= C_S\cdot i(S_2t_p + S_1t_s)\begin{bmatrix}-(1-\cos\theta_d)\cos\phi\sin\phi\\ \cos^2\phi + \sin^2\phi\cos\theta_d\end{bmatrix}\\ &= \frac{1}{2}C_S\cdot i(S_2t_p + S_1t_s)\begin{bmatrix}-(1-\cos\theta_d)\sin 2\phi\\ (1+\cos\theta_d) + (1-\cos\theta_d)\cos 2\phi\end{bmatrix},\end{aligned} \tag{S6}$$

where t_p and t_s are Fresnel transmission coefficients at the glass and sample/water interface for p- and s-polarized light, and C_S is the overall numerical pre-factor including $\eta = (1/\pi)\sin^{-1}(\min(\mathrm{NA}/n_s,1))$, collection efficiency of the objective lens along the angular extent of aperture.

B. Electric fields at the focal region of the imaging lens derived from the Richards-Wolf integral

The electric field at the focal region of an imaging lens is calculated by the R-W integral:

$$\vec{E}(\vec{r}) = -\frac{ik}{2\pi}\int_0^{\theta_{d,\max}}\int_0^{2\pi}\vec{E}_\infty(\theta_d,\phi)e^{ik\vec{s}\cdot\vec{r}}e^{ik\Lambda_{\mathrm{OPD}}}f(\theta_d)\sin\theta_d\,\mathrm{d}\phi\mathrm{d}\theta_d\,, \tag{S7}$$

where the apodization function is given by $f(\theta_d) = \sqrt{\frac{n_i\cos\theta_d}{n_a\cos\theta}}$, $\vec{s}\cdot\vec{r} = (\sin\theta_d\cos\phi\,,\sin\theta_d\sin\phi\,,\cos\theta_d)\cdot(r\cos\phi_d\,,r\sin\phi_d\,,z_d) = r\sin\theta_d\cdot\cos(\phi-\phi_d) + z_d\cos\theta_d$, $\theta_{d,\max}$ the semi-aperture angle of θ_d, $k = \frac{2\pi n_a}{\lambda}$ the angular wavenumber of light, n_a the refractive index of air, $r = \sqrt{x^2+y^2}$ the distance from the focus to an observation point, and $\phi_d = \tan^{-1}(s_y/s_x)$ the azimuthal angle of an imaging ray. $\vec{E}_{R,\infty}(\theta_d,\phi)$ and $\vec{E}_{S,\infty}(\theta_d,\phi)$ are inserted in Eq. (S7) in the place of $\vec{E}_\infty(\theta_d,\phi)$ to get $\vec{E}_R$ and $\vec{E}_S$, respectively. As a ray propagates through a series of optical systems, the (polar) angle made by the ray with respect to the optic axis changes. To compute the electric field via the R-W integral, the

polar angle (θ_d) defined with respect to the focus of the imaging lens is converted to the polar angle (θ) defined with respect to the focus of the objective lens by using Snell's law and equation of magnification, $n_i \sin\theta = M n_a \sin\theta_d$, where M is the effective magnification of a microscope [3]. For an imaging system with large M, $\sin\theta_d$ can be approximated to θ_d because the semi-aperture angle of the imaging lens in such a system is very small. Thus, the following angular relations are useful in the above equation: $d\theta_d = (n_i/M)\cos\theta d\theta$ and $\sin\theta_d d\theta_d = (n_i/M)^2(\sin 2\theta/2)d\theta$ where n_a is set to be 1 [4]. In our microscope setup, the magnification of the objective lens is 62 if the tube lens with a focal length of 180 mm is used as recommended. In fact, we used a tube lens with a focal length of 500 mm and thus, the effective magnification of our microscope is 170.

$$\vec{E}(r) = -\frac{ik}{2\pi}\int_0^{\theta\max}\int_0^{2\pi}\vec{E}_\infty(\theta_d,\phi)e^{ikz_d\cos\theta_d}e^{ikr\sin\theta_d\cos(\phi-\phi_d)}e^{ik\Lambda_{\mathrm{OPD}}}\sqrt{\frac{n_i\cos\theta_d}{n_a\cos\theta}}\left(\frac{n_i}{M}\right)^2\left(\frac{\sin 2\theta}{2}\right)\mathrm{d}\phi\mathrm{d}\theta\,, \quad \text{(S8)}$$

Where $\theta_{\max}$ is the semi-aperture angle of θ and related to $\theta_{d,\max}$ by $n_i\sin\theta_{\max} = M n_a \sin\theta_{d,\max}$.

To simplify this integral, the azimuthal angle (ϕ) is integrated first by using the following relations:

$$\int_0^{2\pi}\cos n\phi\, e^{ix\cos(\phi-\varphi)}\mathrm{d}\phi = 2\pi(i^n)J_n(x)\cos n\varphi\,, \quad \text{(S9)}$$

$$\int_0^{2\pi}\sin n\phi\, e^{ix\cos(\phi-\varphi)}\mathrm{d}\phi = 2\pi(i^n)J_n(x)\sin n\varphi\,, \quad \text{(S10)}$$

The electric field is then simplified as follows:

$$\vec{E}(r) = -\frac{ik}{2}\left(\frac{n_i}{M}\right)^2\int_0^{\theta_{\max}}\vec{E}_a e^{ikz_d\cos\theta_d}e^{ik\Lambda_{\mathrm{OPD}}}\sqrt{\frac{n_i\cos\theta_d}{n_a\cos\theta}}\sin 2\theta\mathrm{d}\theta\,, \quad \text{(S11)}$$

where the strength vectors $\vec{E}_a$ for reference and scattered fields are

$$\vec{E}_{a,\mathrm{ref}} = \frac{1}{2}C_R\cdot i(r_p + r_s)\begin{bmatrix}(1-\cos\theta_d)J_2(kr\sin\theta_d)\sin 2\phi_d \\ (1+\cos\theta_d)J_0(kr\sin\theta_d) - (1-\cos\theta_d)J_2(kr\sin\theta_d)\cos 2\phi_d\end{bmatrix}, \quad \text{(S12)}$$

$$\vec{E}_{a,\mathrm{scat}} = \frac{1}{2}C_S\cdot i(S_2t_p + S_1t_s)\begin{bmatrix}(1-\cos\theta_d)J_2(kr\sin\theta_d)\sin 2\phi_d \\ (1+\cos\theta_d)J_0(kr\sin\theta_d) - (1-\cos\theta_d)J_2(kr\sin\theta_d)\cos 2\phi_d\end{bmatrix}, \quad \text{(S13)}$$

where $\sin\theta_d = (1/M)n_i\sin\theta$ and $\cos\theta_d = \sqrt{1-(n_i\sin\theta/M)^2}$.

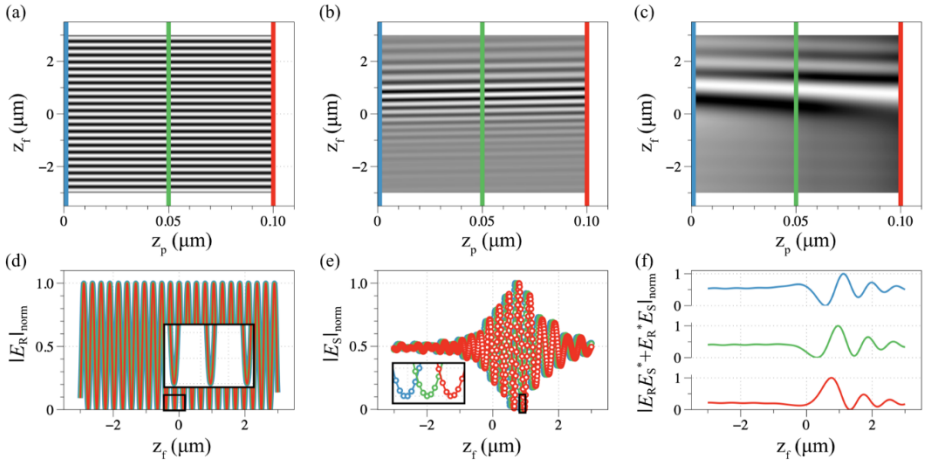


Fig. S3. Stacked axial contrast profile as a function of the particle height (z_p). (a) Reference field ($|E_R|$), (b) scattered field ($|E_S|$) and (c) interference term ($|E_R E_S^* + E_R^* E_S|$) calculated in the range of $z_p = 0 \sim 100$ nm with $n_i = 1.51$. All calculated values are scaled to fit to [0,1] and displayed in a gray scale. The blue, green, and red colored lines indicate $z_p = 0$, 50, and 100 nm, respectively. (d-f) Cross-sections of the reference field (d), the scattered field (e), and the interference term (f). Insets in (d) and (e) show the phase variation of the fields and highlight the sensitivity of the fields to different values of z_p. The axial intensity profile by the interference term exhibits more pronounced variation with z_p accounting for sensitive detection of the size of nano-objects by iSCAT. In this figure (a-f), z_f is the computational input parameter before the offset (δz) is adjusted (see main text for details).

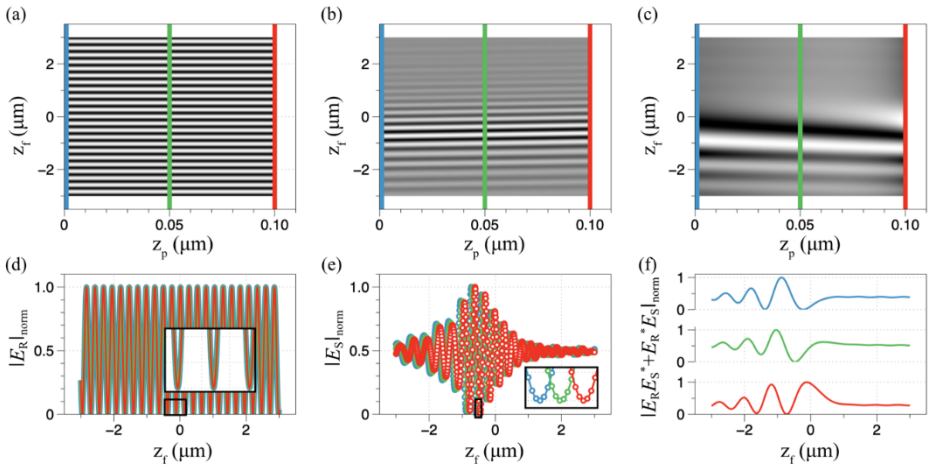


Fig. S4. Simulations performed and results displayed identically to Fig. S3 except for n_i = 1.53.

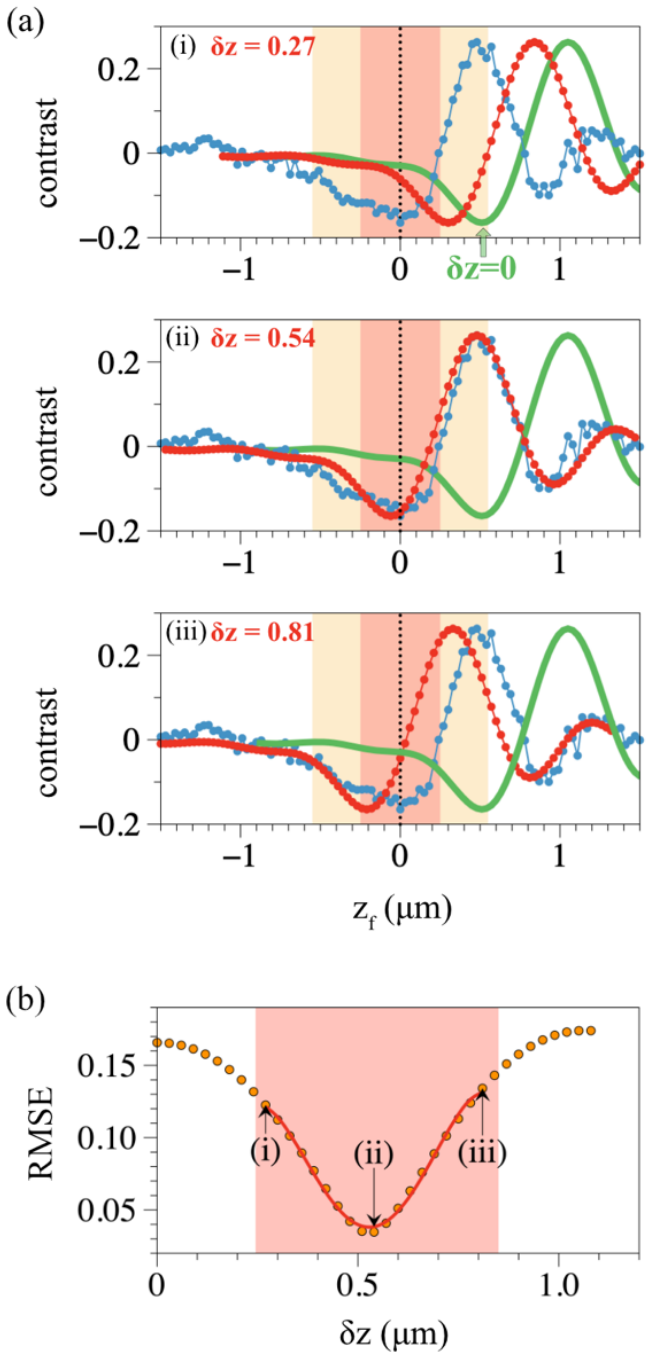


Fig. S5. Root-mean-squared-error (RMSE) estimation to find the best theoretical axial intensity profile (I_{sim}) with a given set of n_i and z_p as a function of the relative focal position (z) that matches the experimental (I_{exp}) profile with the highest similarity. (a) Demonstration of the profile matching procedure via translation of the simulated profile (I_{sim}) along z_f by δz. The RMSE value is computed from I_{exp} (blue line with dots) and I_{sim} (red line with dots) over the shaded region with good overlap. (b) RMSE values as a function of δz, which is the offset applied to I_{sim}. Once the value of δz to yield the least RMSE is found, the offset is denoted as δz^*. (i) $\delta z = 0.27$ μm, (ii) $\delta z = 0.54$ μm = δz^*, and (iii) $\delta z = 0.81$ μm with respect to the $\delta z = 0$ μm (green solid line in (a)). We fitted the RMSE values to a 4th polynomial function to determine the representative RMSE for the given set of n_i an z_p.

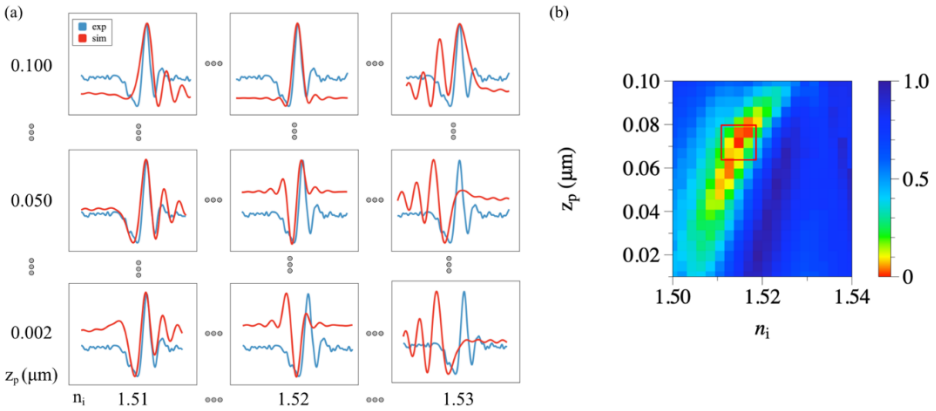


Fig. S6. RMSE estimation to determine the model parameters, n_i and z_p for a 60-nm latex bead. (a) Illustration of the parameter-finding procedure by minimizing the RMSE between the measured axial contrast profile (blue line: experimental) and the computed axial contrast profile over the whole parameter space of n_i and z_p (red line: simulated). (b) Pseudo-color maps of RMSE re-scaled to fit to [0,1] for various n_i and z_p. The minimum RMSE marked by a red box represents the best fit of the PSF modeling for the measured axial contrast profile. The n_i and z_p at the minimum RMSE are used as the parameters to generate the PSF with the highest similarity.

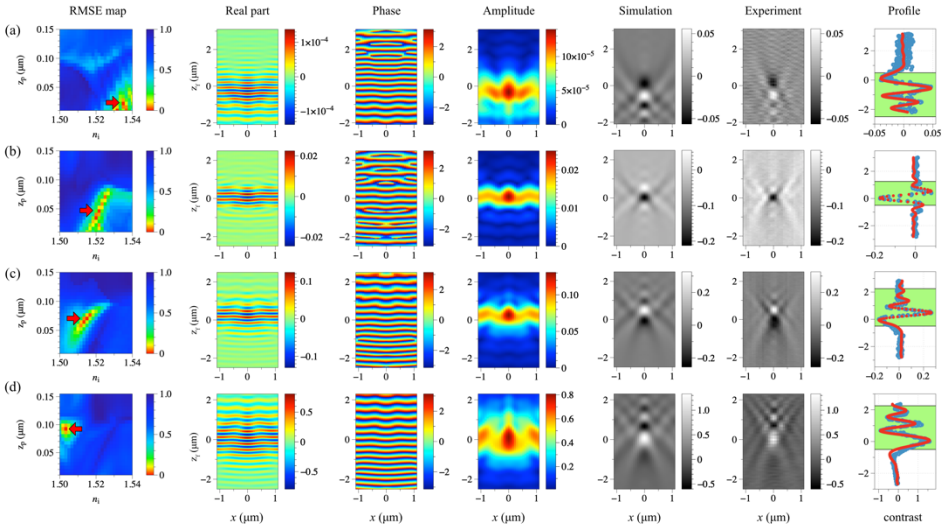


Fig. S7. Representative results of PSF modeling analysis for different-sized dielectric nanoparticles ((a) R_{nom} = 20 nm, (b) 50 nm, (c) 60 nm, and (d) 100 nm). First column: The RMSE map shows the representative RMSE values for the whole parameter space of n_i and z_p, trying to find the best fit (marked by a red arrow). Second column: Real part of the electric field of the scattered light, E_S. Third column: Phase of the electric field of the scattered light, E_S. Fourth column: Amplitude of the scattered light, $|E_S|$. The SHIP images, one by simulation (fifth column) and one by experiment (sixth column) with the best parameter values [(a) (n_i, z_p) = (1.538, 0.022 μm), (b) (n_i, z_p) = (1.519, 0.039 μm), (c) (n_i, z_p) = (1.515, 0.052 μm), (d) (n_i, z_p) = (1.503, 0.093 μm) with design parameters (n_s, n_p, n_i^*, n_g^*, t_i^*, t_g^*) = (1.33, 1.59, 1.52, 1.52, 100 μm, 170 μm)], exhibit clear similarity, indicating the reliability of the parameters obtained here. The axial contrast profile in the seventh column shows the best simulation curve (red line with dots) for each experimental curve (blue line with dots). Green highlighted area defines the iSCAT features for final fitting and error calculation.

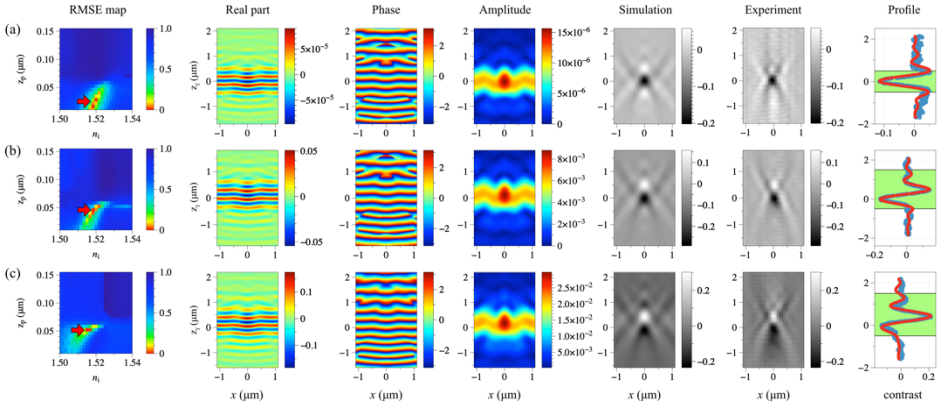


Fig. S8. Representative results of PSF modeling analysis for different-sized fluorescent nanodiamonds (fND) particles from the same product number (900172, Sigma-Aldrich). This figure is presented in the same manner as Figure S7. First column: The RMSE map shows the representative RMSE values for the whole parameter space of n_i and z_p, trying to find the best fit (marked by a red arrow). Second column: Real part of the electric field of the scattered light, E_S. Third column: Phase of the electric field of the scattered light, E_S. Fourth column: Amplitude of the scattered light, $|E_S|$. The SHIP images, one by simulation (fifth column) and one by experiment (sixth column) with the best parameter values [(a) (n_i, z_p) = (1.521, 0.012 μm), (b) (n_i, z_p) = (1.518, 0.045 μm), (c) (n_i, z_p) = (1.516, 0.055 μm), with design parameters (n_s, n_p, n_i^*, n_g^*, t_i^*, t_g^*) = (1.33, 2.4, 1.52, 1.52, 100 μm, 170 μm)], exhibit clear similarity, indicating the reliability of the parameters obtained here. It is also notable that the size of fND is highly heterogeneous. The axial contrast profile in the seventh column shows the best simulation curve (red line with dots) for each experimental curve (blue line with dots). Green highlighted area defines the iSCAT features for final fitting and error calculation.

References

1. J. Kim, Y. Wang, and X. Zhang, “Calculation of vectorial diffraction in optical systems”, J. Opt. Soc. Am., **35**(4), 526-535 (2018).
2. C. F. Bohren and D. R. Huffman, *Absorption and Scattering of Light by Small Particles* (Wiley, 1983).
3. J. Dong, D. Maestre, C. Conrad-Billroth and T. Juffmann, “Fundamental bounds on the precision of iSCAT, COBRI and dark-field microscopy for 3D localization and mass photometry”, Appl. Phys., **54**, 394002 (2021).
4. L. Novotny, R. D. Grober, and K. Karrai, “Reflected image of a strongly focused spot”, Opt. Lett., **26**(11), 789-791 (2001).